

Prtf1 Regex and OpenRefine

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Task 1: List regex that extracts dates from <http://bit.ly/regexexercise2> and formats them as YYYY-MM-DD?

URL for the solution: <https://regex101.com/r/EYyEu9/1>

Brief explanation: Regular expression can be used to search for dates and names. To extract all the dates and put them into the format of YYYY-MM-DD, we divided the date, month and year into three separate groups using the regular expression:

`(\d{1,2}).(\d{1,2})..?(\d{4})`

We also experimented with a simplified version of the expression:

`(\d+).(\d+).\s?(\d+)`

We did it in the following order: month, date and year. The dot "." is used because it symbolises every character, and thereby makes the expression flexible, so it can match dates written with different separators such as: DD-MM-YYYY or DD/MM/YYYY. Using parenthesis indicates that each group is different.

After we successfully divided them into categories, we clicked on substitution. In substitution, we could control the three categories by putting them in the correct order: year, month and date with the following input: \$3-\$2-\$1

We are now able to emphasize the dates in the following format: YYYY-MM-DD.

Task 2 Regex to convert stopwords between R and Voyant format:

URL for a regular expression from Voyant into a stopword list for R: <https://regex101.com/r/UjE8uO/1>

URL for stoplist from R to Voyant: <https://regex101.com/r/7r4nzR/2/substitution>

Brief explanation: First we handle the spaces and separators between the words, which is faster and easier. In the substitution, we wrap each word in quotation marks and separate them with commas. We must beware that the first and last word needs one quotation mark each, which needs to be added. To make the stoplist from R to Voyant, we used a regular expression with alternation (*) to match both the separator ", and the remaining quotation marks ". By replacing these matches with a new line, we removed all quotes and turned the commas into line breaks. This converts the R format ("og","i","jeg") into the Voyant format with one word per line.

Task 3 Does OpenRefine alter the raw data during sorting and filtering?

NO

Brief explanation: OpenRefine does not alter the raw data when you sort or filter a dataset. Sorting only changes the order in which the rows are displayed on the screen, and filtering merely hides rows that do not meet the selected criteria. In both cases, the underlying data remains the same. The raw data is only changed when you apply transformations such as editing values, splitting or merging columns, or using other data-cleaning functions.

Task 4: What are the two most water-deprived months in Mozambique and Tanzania?

Work process:

- 1) Create a text facet under "Months_no_water", and select "sort by count"
- 2) Select "Edit cells" → "transform" and transform the following:
 - a. `value.replace("[", "").replace("]", "").replace("'", "")`
- 3) Select "Edit cells" → "split multi-valued cells" and write ";" under "replace"
- 4) Remember to leave the space between values. Select "edit cells" → "transform" and write the following:
 - a. `value.replace(" ", "")`

Result: now it clearly states that the driest months are in fact October(74) and September(70)

Task 5: Challenge - 10 most common occupations among unmarried women in 1801

Aarhus

- Define what is / is not an occupation:

Definition for occupation → An occupation is a regular type of profession and is done to make a living. It is a socially recognized role tied to the economic structure of a society.

Definition for not an occupation → If an activity is done with the purpose of enjoyment or not to make a livelihood, it is not considered an occupation. In a historical context, it could be to have a societal role not performing manual labour, such as "living by her husband's means" or being unable to work.

- 10 most frequent occupations (list number of people practising that job with each entry)

Work process:

- 1) Create a text facet under category "koen", and exclusively include the word "kvinde"
- 2) Create a text facet under category "civilstand" and exclusively include the word "ugift"
- 3) Under category "alder", select "edit cells" → "common transform" → "to number"
- 4) Select the number to only include the ages of 20-30
- 5) Create a text facet under category "erhverv" and cluster relevant occupations together that serve the same purpose such as "interste og væver" and "inderste og vævpige" → use the "key collision" method and the "metaphone 3" key function

1. Tjenestepige (48)
2. Væverske (33)
3. Skrædder (18)
4. Husjomfru (10)
5. Spinderske (10)
6. Husholderske (7)
7. Kokkepige (5)
8. Bryggerpige (4)
9. Daglejer (1)
10. Barnepige (1)